Report: Static Routing Between Two Routers Connecting Two LANs

Tool: Cisco Packet Tracer | **Topic:** Static routing and end-to-end reachability

1. Objective

To configure static routes between two routers that connect two separate LANs, verify full reachability between the LANs, and prove the packet path with a traceroute.

2. Topology

```
PC0, PC1 -- Switch0 -- Router0 ====== Router1 -- Switch1 -- PC2, PC3
     LAN A                    Link                    LAN B
 192.168.1.0/24             10.0.0.0             192.168.2.0/24
```

3. Addressing

Device / Interface	IP Address	Mask
Router0 Gi0/0 (LAN A gateway)	192.168.1.1	255.255.255.0
Router0 Gi0/1 (link to Router1)	10.0.0.1	255.0.0.0
Router1 Gi0/1 (link to Router0)	10.0.0.2	255.0.0.0
Router1 Gi0/0 (LAN B gateway)	192.168.2.1	255.255.255.0
PC0	192.168.1.10	255.255.255.0
PC2	192.168.2.10	255.255.255.0

Mask note: Router0's screenshot shows the router-to-router link configured with 255.0.0.0 (/8). This works because both ends use the same mask. A /30 (255.255.255.252) would be more efficient for a two-device link and is recommended in real designs.

4. Configuration

Router0

```
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 ip address 10.0.0.1 255.0.0.0
 no shutdown
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Router1

```
interface GigabitEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 ip address 10.0.0.2 255.0.0.0
 no shutdown
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Each static route follows the form `ip route <destination network> <mask> <next-hop>`. Router0 reaches LAN B through Router1 (10.0.0.2), and Router1 reaches LAN A through Router0 (10.0.0.1). Both directions are needed, because a reply cannot return without a route back to the source.

5. Verification

5.1 Routing table (Router0, show ip route)

```
C    10.0.0.0/8 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.2
```

- `C` and `L` entries are the directly connected networks and local interface addresses.
- The `S` entry shows the static route to LAN B, with administrative distance 1 and metric 0, via next hop 10.0.0.2.

- Router0 learned LAN B only because of the static route, since it is not directly connected.

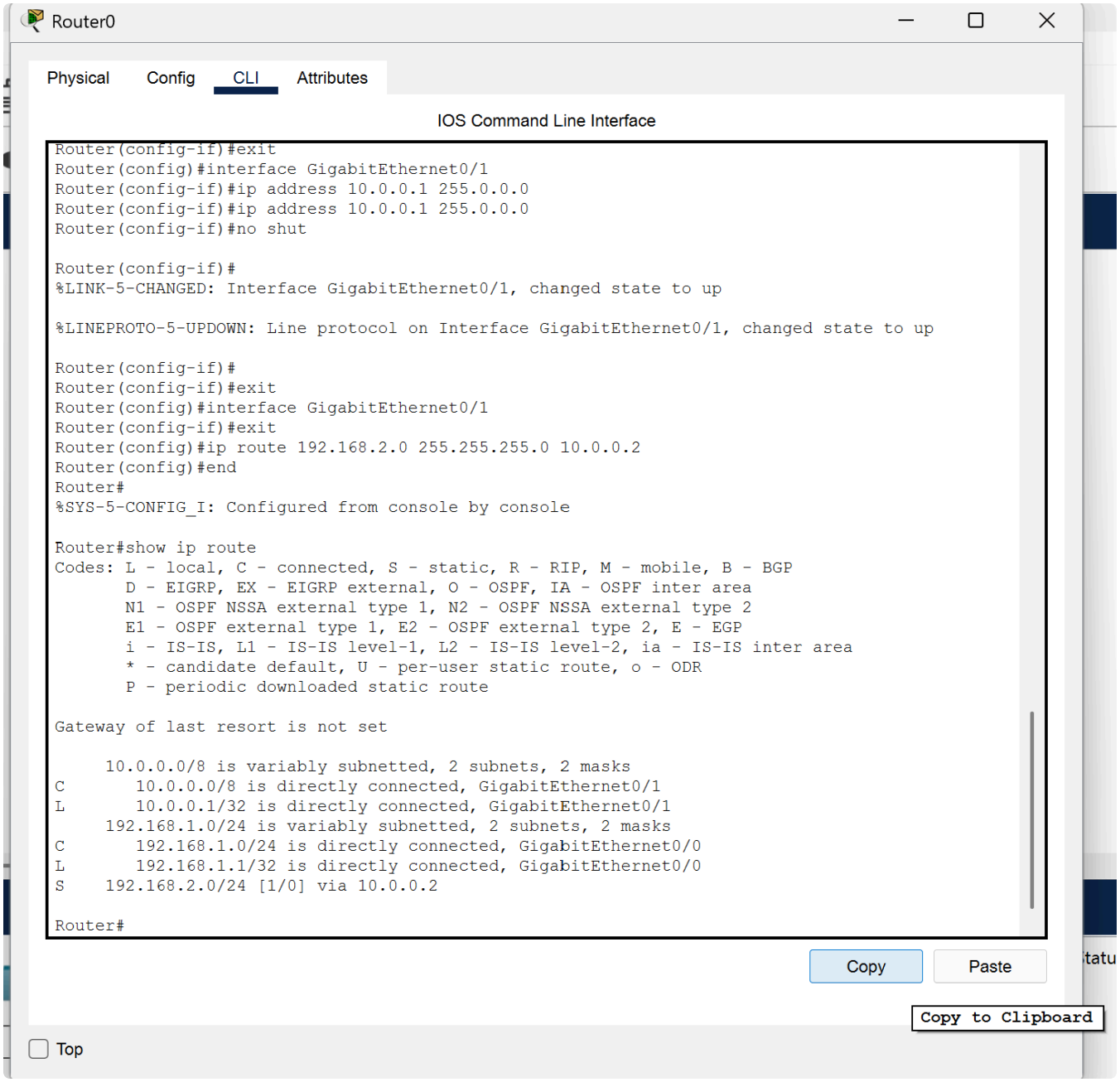


Figure 1: Router0 CLI showing the S route to 192.168.2.0/24 via 10.0.0.2.

5.2 Ping test (PC0 to PC2, 192.168.2.10)

Attempt	Sent	Received	Lost	Result
First ping	4	2	2	50% loss

Attempt	Sent	Received	Lost	Result
Second ping	4	4	0	0% loss

The first two packets timed out because ARP resolution and the initial forwarding setup had not finished. Once the ARP entries were learned, every packet succeeded, with replies in under 1 ms. The reply TTL of 126 confirms the packets crossed two routers (starting TTL 128 minus 2 hops).

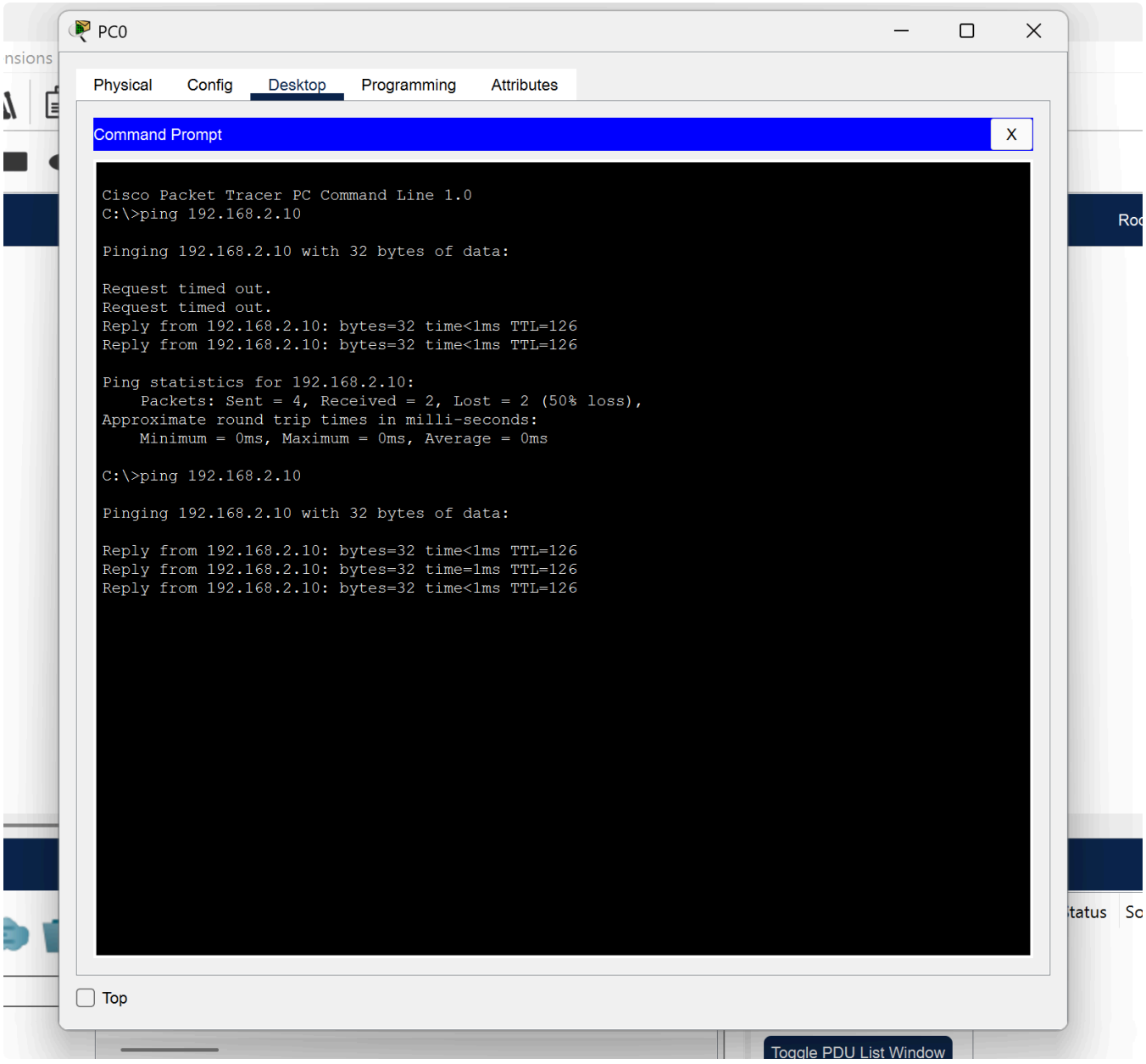


Figure 2: PC0 pinging 192.168.2.10 (first attempt 50% loss from ARP, second attempt succeeds).

5.3 Traceroute (PC0 to PC2, tracert 192.168.2.10)

```
1    0 ms    0 ms    0 ms    192.168.1.1      (Router0, LAN A gateway)
2    0 ms    0 ms    0 ms    10.0.0.2         (Router1, link interface)
3    0 ms    0 ms    0 ms    192.168.2.10     (PC2, destination)
Trace complete.
```

This proves the path: **PC0 → Router0 → Router1 → PC2**. Traffic leaves LAN A through Router0's gateway, crosses the 10.0.0.0 link to Router1, and is delivered to the host in LAN B. This is the path defined by the static routes.

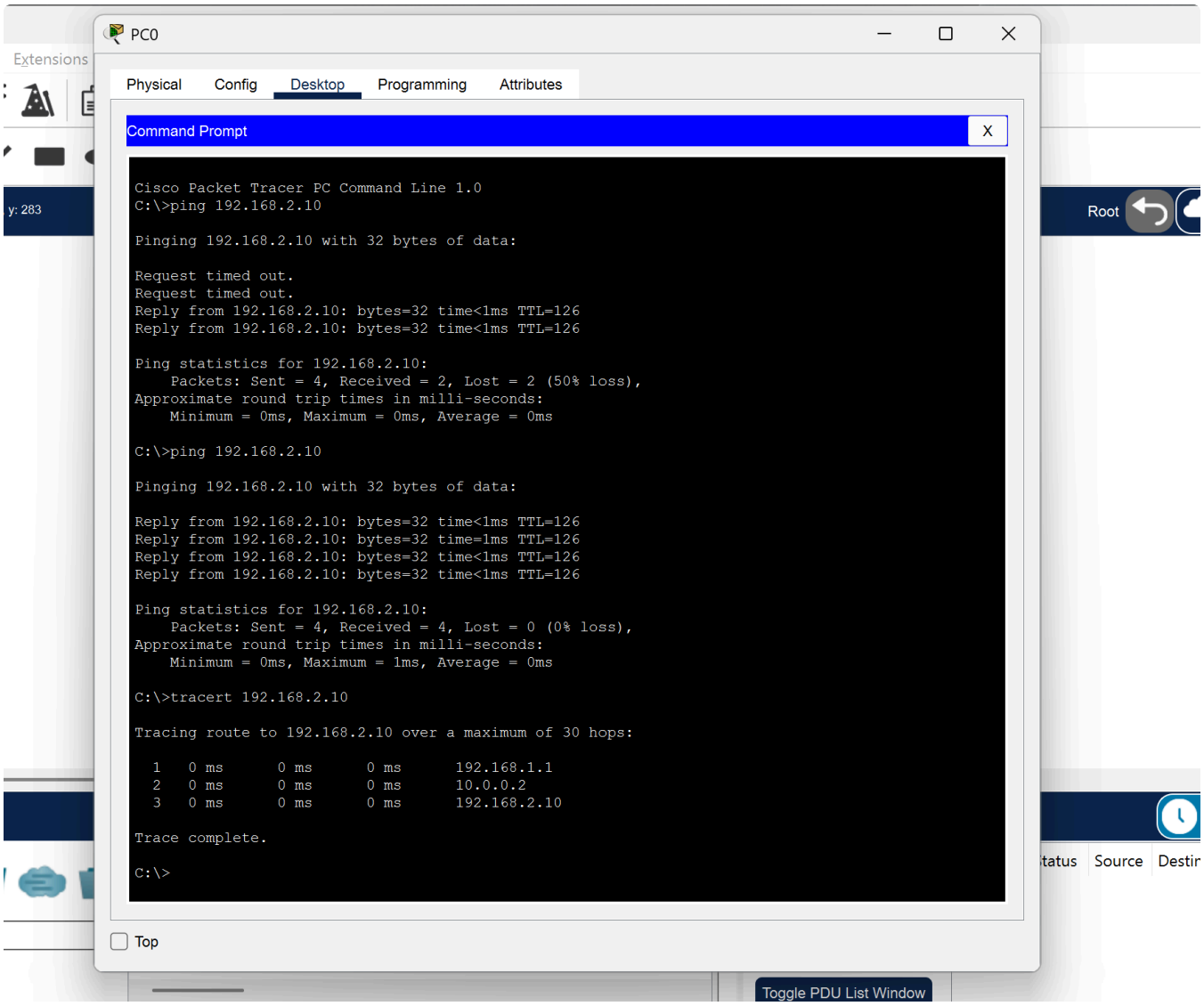


Figure 3: PC0 ping results and tracert showing the path 192.168.1.1, 10.0.0.2, 192.168.2.10.

6. Results

Test	Outcome
Interfaces up/up	Passed
Static route in routing table (S)	Passed
LAN A to LAN B ping	Passed (0% loss after ARP)
Traceroute shows expected path	Passed

7. Conclusion

Static routes on both routers allowed two separate LANs to communicate. Without them, each router knew only its directly connected networks and inter-LAN traffic failed. After adding the routes, `show ip route` displayed the `S` entry, pings succeeded, and the traceroute showed the exact hop-by-hop path through both routers.

Advantages of static routing: simple, no routing protocol overhead, predictable paths, and low CPU and bandwidth use.

Limitations: it does not scale to large networks, requires manual updates, and cannot reroute around link failures automatically. Dynamic protocols such as OSPF handle those cases.

8. Attachments

1. Screenshot: Router0 CLI showing `show ip route` (the `S` route)
2. Screenshot: PC0 ping to 192.168.2.10
3. Screenshot: PC0 `tracert 192.168.2.10`
4. Router1 `show running-config` and `show ip route` (add these, as only Router0 was in the original screenshots; Router1 should show `S 192.168.1.0/24 [1/0] via 10.0.0.1`)